Visualisations and Sample Outputs

MAAI901 Deep Learning for Visual Data, Assessment 2: Locating Photos by Country. Ahmad Hello, 7573297. 27 October 2026.

1. Results

1.1 Main test results (%, mean of three seeds)

Model	Top-1 (mean ± SD)	Macro-F1	Head only (stage 1)	Unseen contributors	Urban	Natural
CLIP ViT-B/16	75.5 ± 0.1	75.2	64.4	53.8	74.9	72.1
ViT-B/16 (ImageNet-21k)	64.3 ± 0.1	63.8	34.2	37.5	58.9	60.8
ResNet-50	62.1 ± 0.1	61.6	30.8	35.9	55.7	58.6

Urban and natural accuracy are averaged over the 21 countries with at least 50 photos of each type. Unseen contributors: the 3,729 test photos whose contributor is not in the training sample.

1.2 Every final run

Model	Test top-1, seed 0	Seed 1	Seed 2	Best validation, seed 0	Seed 1	Seed 2
CLIP ViT-B/16	75.5	75.6	75.5	85.9 (epoch 8)	85.9 (epoch 8)	85.9 (epoch 8)
ViT-B/16 (ImageNet-21k)	64.3	64.2	64.5	75.5 (epoch 8)	75.0 (epoch 8)	74.6 (epoch 8)
ResNet-50	62.2	62.2	62.0	72.6 (epoch 7)	72.2 (epoch 8)	73.1 (epoch 7)

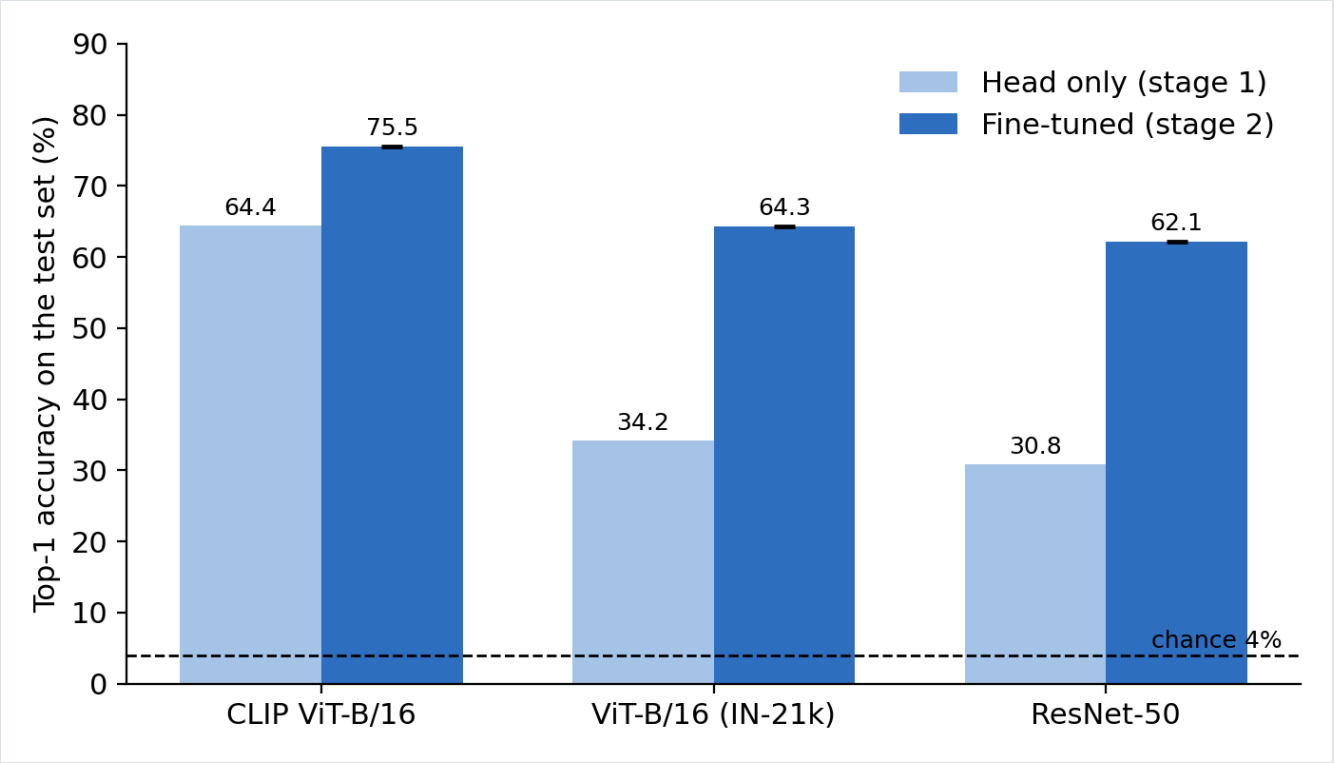


Figure 1. Test accuracy after stage 1 (head only, light bars) and stage 2 (full fine-tuning, dark bars). Error bars on stage 2 show the standard deviation over seeds. IN-21k: ImageNet-21k.

1.3 Statistical tests

Comparison	Seed	Only first model right	Only second model right	McNemar p
CLIP vs ImageNet-21k ViT	0	4,051	1,261	< 0.001
CLIP vs ImageNet-21k ViT	1	4,107	1,244	< 0.001
CLIP vs ImageNet-21k ViT	2	4,097	1,350	< 0.001
ImageNet-21k ViT vs ResNet-50	0	2,686	2,171	< 0.001
ImageNet-21k ViT vs ResNet-50	1	2,711	2,203	< 0.001
ImageNet-21k ViT vs ResNet-50	2	2,780	2,155	< 0.001

1.4 Training

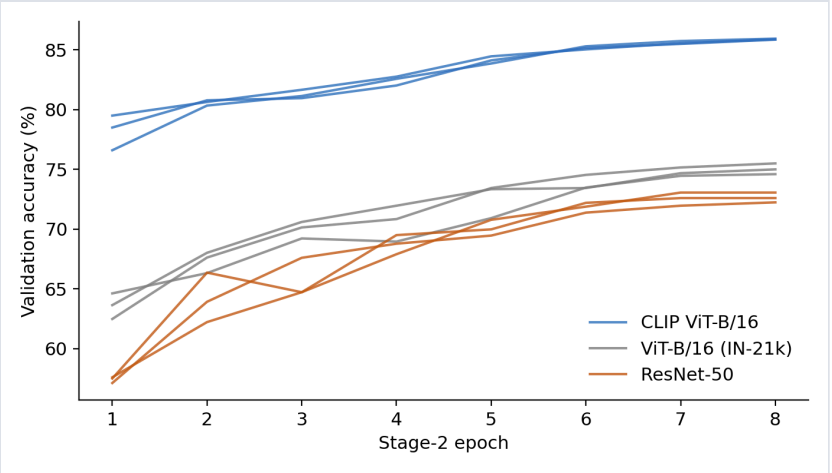


Figure 2. Validation accuracy in each stage-2 epoch for the nine final runs (one line per model and seed). The best epoch was always the 7th or 8th.

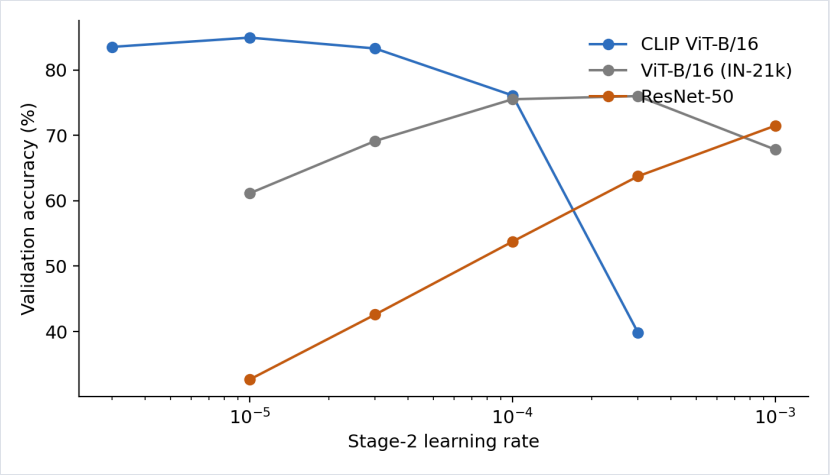


Figure 3. Learning-rate search (seed 0, shortened 3-epoch stage 2): best validation accuracy at each rate. Chosen rates: 10^{-5} for CLIP, 3×10^{-4} for the ImageNet-21k ViT and 10^{-3} for ResNet-50.

1.5 Urban and natural scenes (H2)

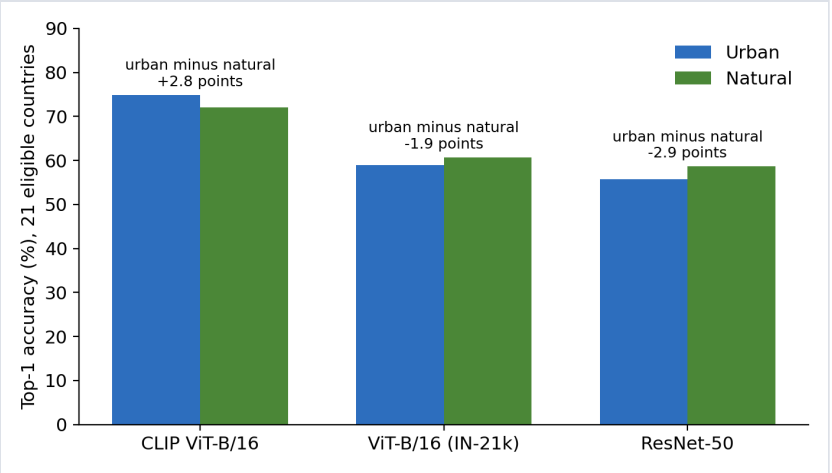


Figure 4. Urban and natural accuracy, averaged over the 21 eligible countries. Only CLIP was less accurate on natural scenes.

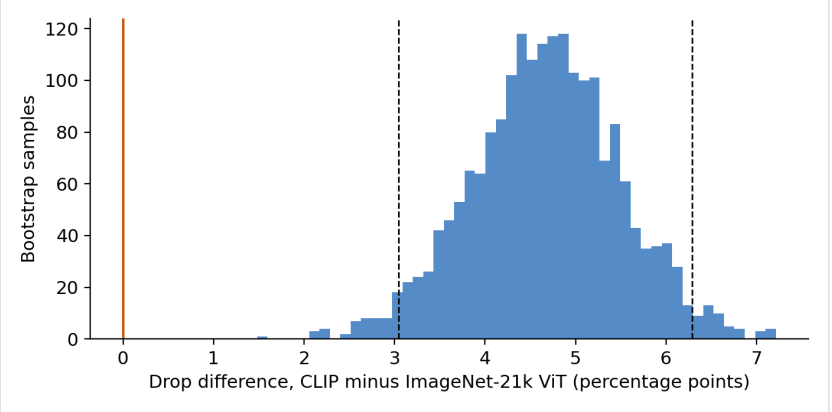


Figure 5. Bootstrap distribution (2,000 resamples of whole sequences) of the difference between CLIP's and the ImageNet-21k ViT's urban-to-natural drop. Difference 4.7 points, 95% interval 3.0 to 6.3 (dashed lines); zero (orange line) lies outside it. With confident scene labels only: 4.8 points, interval 2.7 to 6.9.

1.6 Countries and confusions (RQ3)

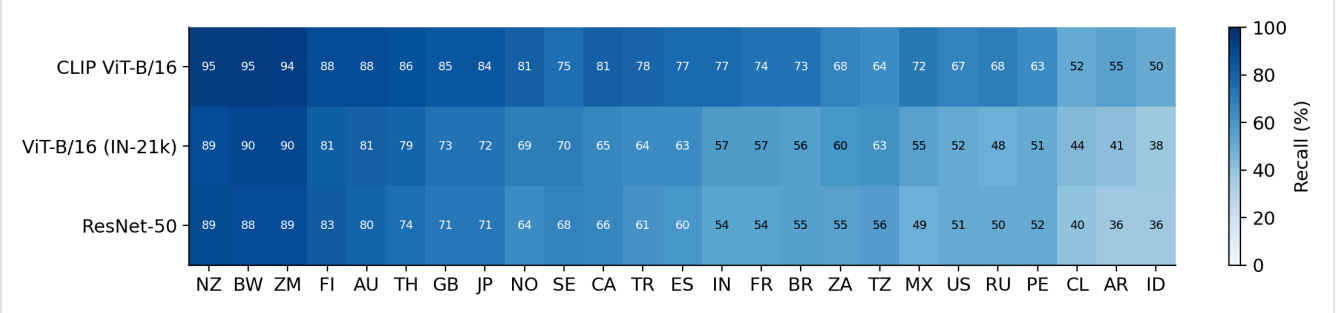


Figure 6. Recall (%) per country and model, sorted by the three-model average (ISO country codes).

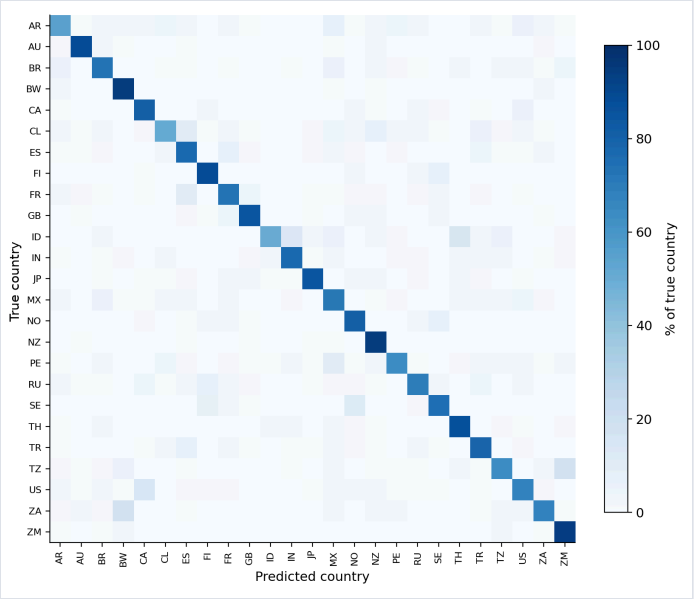


Figure 7. Confusion matrix of CLIP ViT-B/16, all three seeds pooled (rows: true country; each row sums to 100%).

True country	Predicted as	Photos (3 seeds pooled)
Tanzania	Zambia	550
South Africa	Botswana	498
Indonesia	Thailand	484
United States	Canada	441
Indonesia	India	395
Sweden	Norway	344

CLIP's most common mistakes. 17% of CLIP's errors fell inside the six look-alike groups set in advance, against about 3% if errors were spread evenly.

2. Sample outputs

2.1 Model predictions on test photos

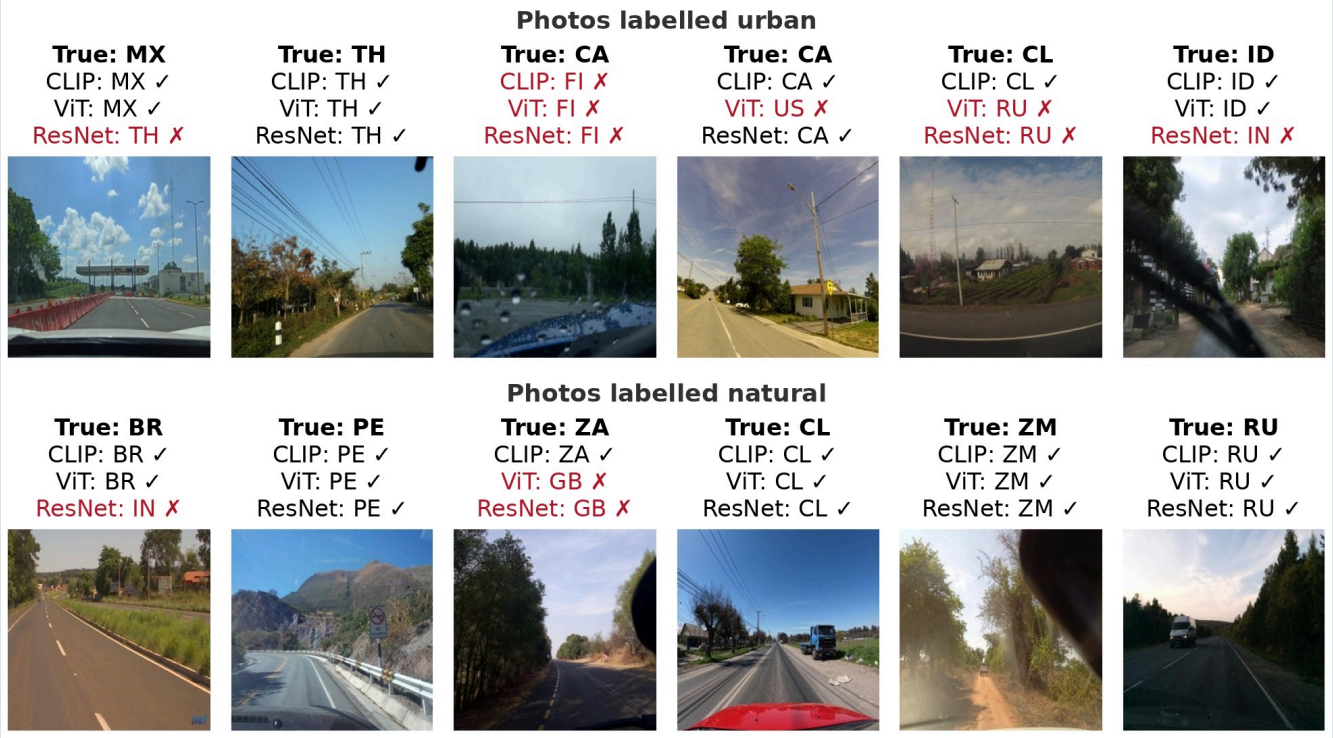


Figure 8. Seed-0 predictions on 12 random test photos (red = wrong, ViT = ImageNet-21k ViT). CLIP got 11 right, ViT 8 and ResNet-50 6. Images by Mapillary contributors, CC BY-SA.

2.2 Scene classifier outputs, checked by eye

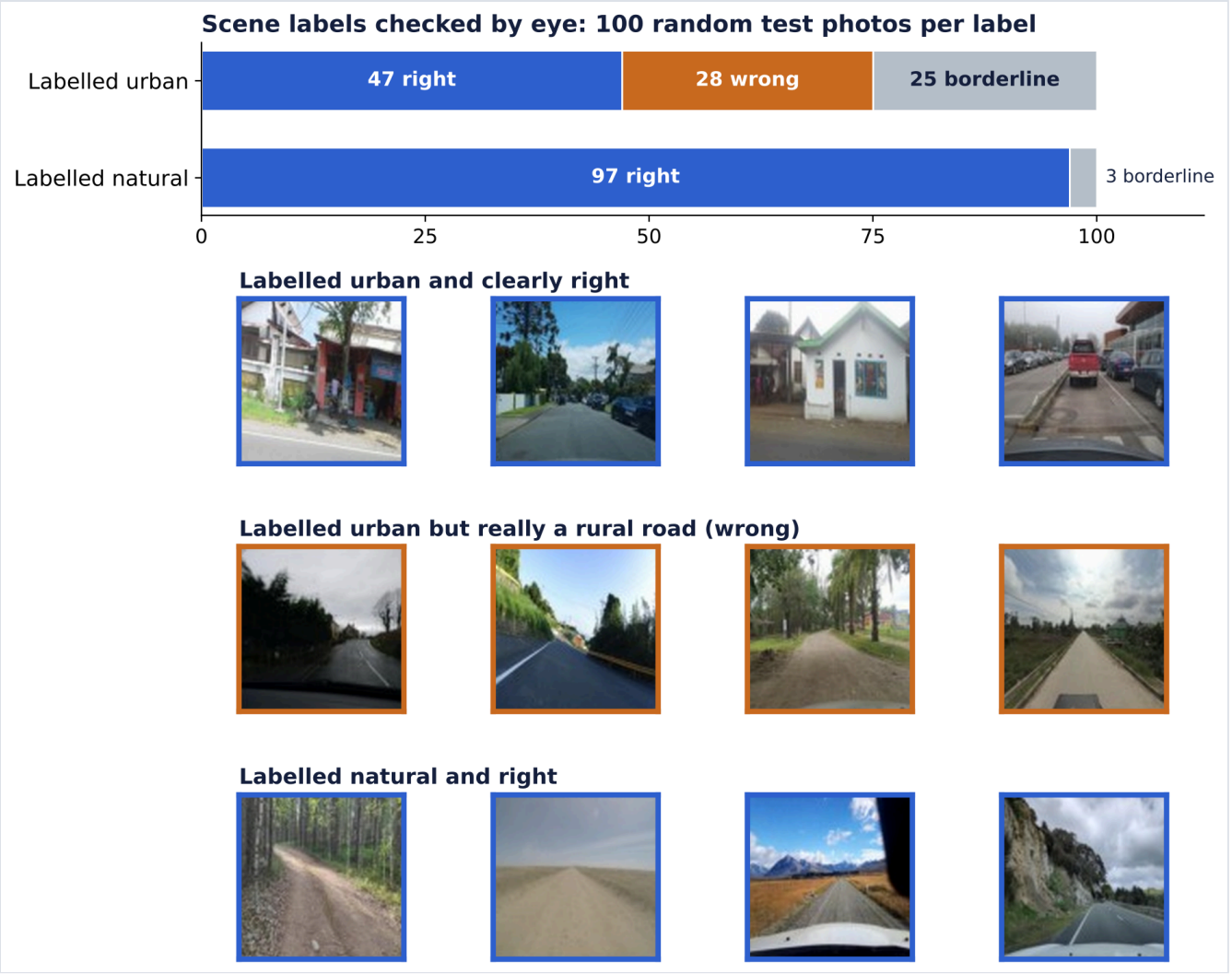


Figure 9. Scene labels of 100 random test photos per label, checked by eye, with four examples each of right urban, wrong urban and right natural labels. Natural labels were almost always right, but many urban labels were really rural roads.